# Quantum Circuit Optimization Using Geometric Programming: Convex Duration Allocation on Fixed NISQ Circuits

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***Abstract*—** We study gate-duration allocation on a fixed quantum-circuit dependency graph under a declared inverse-duration plus exposure-error model. The continuous problem admits geometric-program and ordinary convex formulations; a path-flow dual bounds its unconstrained variant. For Clifford circuits with independent isotropic Pauli channels, backward Pauli tracking yields convex stabilizer-expectation constraints within the model, and a geometric-grid rule preserves feasibility under a certified safety margin. In a saved IBM calibration-snapshot study, a model-attainable 24-instance protocol demonstrates continuous solves through 128 qubits (median solve time 6.41 s at 128 qubits). Across 64 and 128 qubits, median modeled makespan reduction against an optimal uniform-duration baseline plateaus identically at 24.5%; we prove analytically that this plateau arises from boundary saturation of the duration box $[\ell, u] = [0.5, 3.0] \times \tau_{\text{base}}$ as the circuit depth-to-gate ratio stabilizes. In matched-deadline multi-circuit simulations across 48 benchmark circuits in three drive regimes, average process fidelity changes range from −0.0096 pp (Low Drive, with 17 regressions out of 47 feasible instances and one solver infeasibility) to +0.0258 pp (Balanced), with median gains of exactly 0.0000 pp across all regimes. Under structural noise misspecification (quadratic $1/\tau^2$ error), GP durations perform strictly worse than uniform schedules in 100% of evaluated circuits. Direct comparisons against Qiskit scheduling show that Markovian gains ($\sim 3\times10^{-4}$) are largely self-predictions of the aligned model, while unmodeled shot-static detuning induces severe rank reversals favoring dynamical decoupling (0.9934 vs 0.9382). Finally, a physical experiment on IBM Heron comparing native 36-ns and 1,000-ns X gates shows that duration-aware choices alter decoherence dynamics, but reveals observable-dependent rank reversals (+4.25 pp in the X basis, −4.57 pp in the Z basis) and thin statistics (three blocks on one qubit) that challenge isotropic single-scalar error models. These findings establish the mathematical limits and modeling boundaries of convex timing, demonstrating that physical efficacy remains contingent on calibrated, anisotropic device models.

***Index Terms*—** Quantum compilation, geometric programming, convex optimization, gate scheduling, Markovian noise, pulse libraries, Pauli tracking, Heron architecture.

## I. Introduction

QUANTUM compilation connects abstract circuits to executable pulses and schedules. Gate count, depth, makespan, and noisy-state fidelity are distinct quantities. Shorter schedules can demand stronger control and may increase error. We therefore study a precise subproblem: duration allocation on a fixed dependency graph under a declared noise model. Its optimum and error bounds are conditional on that model, rather than guarantees about an uncharacterized device.

The mathematical structure of time-cost tradeoffs on directed acyclic dependency graphs (DAGs) traces back to the foundational network-flow duality of Kelley [1] and Fulkerson [2]. Geometric programming (GP) provides an established positive-variable optimization language with an exact convex transformation under logarithmic change of variables [3]. Its historical application to digital CMOS circuit sizing and timing closure played a foundational role in classical electronic design automation (EDA) [4]. In quantum computing, when gate durations are fixed, makespan minimization reduces to a longest-path computation on a DAG. However, physical quantum devices operate under a different decision space: continuous gate durations and discrete pulse implementations that preserve target local unitaries while trading off drive strength, environmental decoherence, and circuit latency.

Our contribution is a model-based timing method, rather than a claim of optimal circuit synthesis or hardware performance. First, we formulate fixed-DAG duration allocation as convex optimization and derive a path-flow lower bound. Second, under an explicitly post-gate isotropic Pauli channel model, we combine Clifford observable tracking with duration optimization and a conditional quantization guarantee. Third, we distinguish model-consistency checks from external validation, report matched-deadline comparisons, and supply an additional feasible large-circuit solve study. The inverse-duration coefficient is not inferred from the saved backend snapshot: its share is swept as an assumption. Claims of physical advantage therefore require measured duration–error curves and hardware-compatible controls.

Sections II–VII describe prior work, the mathematical model, its bounds, and the observable formulation. Section VIII reports local simulations, including the feasible large-circuit study. Section IX describes the analysis pass and limitations; Section X concludes.

## II. Related Work and Scientific Context

### A. Duration-Aware Quantum Scheduling

Lussi, de Santiago, and Duzzioni recently introduced duration-aware quantum scheduling by assigning fast pulse implementations to critical-path operations and robust, longer implementations to available off-critical slack [5]. Our work builds upon this insight by formulating continuous duration allocation as an exact convex program, deriving analytical primal-dual optimality certificates, and incorporating nonunitary open-system channel bounds. In quantum compiler systems, noise-adaptive mapping and crosstalk-aware scheduling have been investigated by Murali et al. [12], [13], while pulse-efficient transpilation was developed by Earnest et al. [14]. Dynamical decoupling techniques for open quantum systems have been developed to mitigate low-frequency non-Markovian noise during idle intervals [17], [19].

### B. Convex Scheduling and Classical Foundations

Convex project time-cost optimization on DAGs originates in the classic network-flow duality of Kelley [1] and Fulkerson [2], subsequently generalized to convex cost dual network flow by Ahuja et al. [6]. Geometric programming supplies an exact convex representation when all cost terms are positive monomials or posynomials [3]. For the proposed inverse-duration-plus-duration costs, the original variables also yield an ordinary convex formulation, establishing that efficiency stems from convexity rather than logarithmic change of variables alone.

### C. Noise Modeling and Stabilizer Observables

The no-jump branch of a quantum-trajectory representation underlies the channel certificate [7]. Conditional-gradient methods provide established optimization-gap machinery [8]. Compositional robustness for quantum programs has been formalized by Hung et al. [10], while robust quantum control via sequential convex programming was investigated by Kosut et al. [11]. Sparse Pauli-Lindblad noise characterization has been established by van den Berg et al. [15]. Recently, P. Aigner et al. [16] developed analytical and compressed simulation techniques for noisy stabilizer circuits. Our framework leverages duration-dependent noise sensitivity to convert analytical stabilizer tracking into a posynomial constraint, enabling convex interior-point optimization for compiler scheduling.

## III. Continuous Duration Model and Exact Convexity

Let a native circuit contain $m$ unitary gates and a dependency DAG $\mathcal{D}$ with $r$ edges. Per-qubit order is fixed by the logical circuit; concurrent operations on overlapping qubit supports are forbidden. Gate $i$ has continuous duration $\tau_i \in [\ell_i, u_i]$, with $0 < \ell_i \le u_i$, generated by a Hermitian Hamiltonian $H_i = K_i/\tau_i$ such that

$$e^{-\mathrm{i}K_i} = e^{\mathrm{i}\phi_i} U_i, \quad K_i = \theta_i P_i/2, \tag{1}$$

where $P_i$ denotes the generating Hamiltonian. On superconducting transmon processors such as IBM Heron, native operations comprise single-qubit rotations (SX, X, and virtual RZ) and two-qubit Controlled-Z (CZ) entangling interactions; entangling gates like RXX decompose into native CZ and single-qubit rotations. Because $K_i$ is fixed, the ideal local unitary is duration-independent.

Define the integrated squared control effort $E(\tau) = \sum_i \kappa_i^2/\tau_i$, where $\kappa_i = \|K_i\|_2$. For a schedule with makespan $M$, the integrated noise hazard satisfies

$$\Lambda \le R(\tau, M) = bM + \sum_i \left( \frac{a_i}{\tau_i} + \gamma_i \tau_i \right), \tag{2}$$

where $b, a_i, \gamma_i > 0$. The inverse-plus-linear term $a_i/\tau_i + \gamma_i\tau_i$ represents a stylized phenomenological noise model capturing the fundamental engineering tension between drive-power-induced errors (leakage, non-adiabaticity, and amplifier distortion at small $\tau_i$) and environmental decoherence exposure ($T_1, T_2$ relaxation and dephasing at large $\tau_i$).

The objective balances noise hazard, makespan, and control effort:

$$J(\tau, M) = w_R \frac{R}{R_0} + w_M \frac{M}{M_0} + w_E \frac{E}{E_0}, \tag{3}$$

subject to precedence constraints $f_i + \tau_j \le f_j$ for $(i \to j) \in \mathcal{D}$, completion constraints $\tau_i \le f_i \le M$, and duration bounds $\ell_i \le \tau_i \le u_i$.

**Theorem 1** (Convexity and Conditional Uniqueness)**.** *Suppose $R_0, M_0, E_0 > 0$, $w_R, w_M, w_E \ge 0$, and the gate coefficients are nonnegative. The fixed-DAG duration-allocation problem in* (3) *is an exact geometric program whenever the positive-variable formulation is used, and is convex in the original durations and finish-time variables. Define $A_i = w_R a_i/R_0 + w_E \kappa_i^2/E_0$. If $A_i > 0$ for every nonfixed duration coordinate, the optimal duration vector is unique. This does not imply unique intermediate finish times.*

*Proof.* Write $B = w_R b/R_0 + w_M/M_0 \ge 0$ and $C_i = w_R\gamma_i/R_0 \ge 0$. Eliminating the finish-time variables at fixed $\tau$ gives the reduced objective

$$F(\tau) = \sum_i (A_i/\tau_i + C_i\tau_i) + B \max_{p \in \mathcal{P}} p^\top \tau.$$

The box is compact, all reciprocal terms are convex on positive durations, and the maximum of path sums is convex. Hence a minimizer exists. If every free-coordinate $A_i > 0$, the Hessian of $\sum_i A_i/\tau_i$ is diagonal with entries $2A_i/\tau_i^3 > 0$

on the free-coordinate subspace. The reduced objective is then strictly convex, so its minimizing duration vector is unique. For the GP representation, positive monomials and posynomials express the objective and precedence inequalities; logarithmic substitution yields convex log-sum-exp inequalities. Fixed coordinates can be eliminated. A zero coefficient or weight may remove strict convexity, so uniqueness is not asserted outside the stated condition. □

**Remark 1** (Objective Formulation and Non-Uniqueness under Pure Makespan Minimization)**.** *Theorem 1 guarantees uniqueness of the optimal duration vector only when the reciprocal error coefficients in the reduced objective function are strictly positive ($w_R > 0$ with $A_i > 0$). In contrast, when the problem is formulated to minimize makespan alone subject to error constraints (as in Algorithm 1), the objective depends exclusively on the critical path length. Gates lying off the critical path possess nonzero slack: their durations can be continuously varied within their slack margins without altering the makespan, provided precedence and observable constraints remain satisfied. Consequently, off-critical duration allocations under pure makespan minimization are non-unique and mathematically arbitrary unless regularized by a strictly convex secondary risk objective.*

**Proposition 1** (Positive Inverse-Power Extension)**.** *For a declared exponent $p > 0$, replace the reduced duration objective by*

$$F_p(\tau) = \sum_i \left(A_i \tau_i^{-p} + C_i \tau_i\right) + B \max_{q \in \mathcal{P}} q^\top \tau, \tag{4}$$

*with $A_i, C_i \geq 0$, $B > 0$, and positive duration bounds. This is a geometric program and is convex in the original durations. If $A_i > 0$ for each free coordinate, the optimal duration vector is unique. The path-flow dual of Theorem 2 remains valid with coordinate response*

$$\tau_i(\rho) = \operatorname{clip}\left[\left(\frac{pA_i}{C_i + \rho_i}\right)^{1/(p+1)}, \ell_i, u_i\right] \tag{5}$$

*when $C_i + \rho_i > 0$; if it is zero and $A_i > 0$, the response is $u_i$.*

*Proof.* Each $A_i \tau_i^{-p}$ is a positive monomial, hence the positive-variable objective and precedence ratios satisfy GP grammar. Its second derivative is $p(p+1)A_i \tau_i^{-p-2} \geq 0$, strictly positive on free coordinates with $A_i > 0$. The maximum of path sums is convex, so strict convexity and uniqueness follow as in Theorem 1. Minimax interchange over the compact duration box and path-flow polytope is unchanged. Differentiating $A_i/t^p + (C_i + \rho_i)t$ gives the displayed clipped minimizer. This extension changes the assumed error law; it does not validate any exponent from hardware data. □

**Proposition 2** (Rectangular Box Robustness)**.** *If each coefficient in* (2) *belongs independently to a declared interval $[a_i^{\min}, a_i^{\max}]$, its maximum over that rectangle is attained at the upper endpoints. Replacing coefficients by those endpoints yields an exact robust GP.*

*Proof.* Each coefficient multiplies a nonnegative term, so $R$ is coordinatewise monotone. The upper endpoint vector belongs to the uncertainty rectangle and attains its maximum for every feasible schedule. □

**Proposition 3** (Finite-Scenario Robust GP)**.** *Let a declared finite set $\mathcal{S}$ contain nonnegative coefficient pairs $(A_{is}, C_{is})$ for a common exponent $p > 0$. Under the same positive duration box, DAG constraints, and optional deadline, the minimax problem*

$$\min_{\tau, M} \left\{ \max_{s \in \mathcal{S}} \sum_i (A_{is} \tau_i^{-p} + C_{is} \tau_i) + BM \right\} \tag{6}$$

*is an exact geometric program in epigraph form. Its optimum is robust only with respect to the* declared *scenarios; it is a physical error bound only if the true coefficients are contained in a justified uncertainty set.*

*Proof.* Introduce $z > 0$ and require $\sum_i (A_{is} \tau_i^{-p} + C_{is} \tau_i)/z \leq 1$ for each $s \in \mathcal{S}$. Every left-hand side is a posynomial, and $z + BM$ is a posynomial objective. The timing and deadline constraints retain GP form. For fixed feasible $\tau$, the smallest feasible $z$ is exactly the maximum scenario error, proving epigraph equivalence. Taking a maximum over finitely many convex scenario errors also proves convexity in the original durations. Nothing in this argument assigns statistical coverage to $\mathcal{S}$. □

**Proposition 4** (Quantization Rounding Certificate)**.** *If a rounded vector $\tau'$ remains within $[\ell, u]$, then*

$$|R(\tau') - R(\tau)| \leq \sum_i \left(\frac{a_i}{\ell_i^2} + \gamma_i + b\right) |\tau_i' - \tau_i|. \tag{7}$$

*Proof.* Every path sum changes by at most $\sum_i |\tau_i' - \tau_i|$, bounding makespan change. The derivative magnitude of $a_i/\tau_i + \gamma_i \tau_i$ on $[\ell_i, u_i]$ is bounded by $a_i/\ell_i^2 + \gamma_i$. Summing yields the result. □

## IV. Path-Flow Dual and Algorithm

Let $\mathcal{P}$ denote the set of all root-to-sink path incidence vectors in $\mathcal{D}$. Makespan is $M(\tau) = \max_{p \in \mathcal{P}} p^\top \tau$. Combining objective coefficients into $A_i, C_i > 0$ and $B > 0$, the unconstrained deadline problem becomes

$$\min_{\ell \leq \tau \leq u} \sum_i \left(\frac{A_i}{\tau_i} + C_i \tau_i\right) + B \max_{p \in \mathcal{P}} p^\top \tau. \tag{8}$$

**Theorem 2** (Path-Flow Dual Bracket)**.** *For any dual flow vector $\rho \in B \operatorname{conv}(\mathcal{P})$, define the dual objective and duration response:*

$$D(\rho) = \sum_i \min_{\ell_i \leq t \leq u_i} \left[\frac{A_i}{t} + (C_i + \rho_i)t\right], \tag{9}$$

$$\tau_i(\rho) = \operatorname{clip}\left(\sqrt{\frac{A_i}{C_i + \rho_i}}, \ell_i, u_i\right). \tag{10}$$

*Then strong duality holds:* $\max_\rho D(\rho) = \min_\tau J(\tau)$. *Moreover, every dual-feasible* $\rho$ *provides an inspectable optimality gap:*

$$0 \le J(\tau(\rho)) - J^* \le BM(\tau(\rho)) - \rho^\top \tau(\rho). \tag{11}$$

*Proof.* The set $\mathcal{P}$ contains every source-to-sink path incidence vector of the fixed DAG; $B\,\mathrm{conv}(\mathcal{P})$ consists exactly of nonnegative path mixtures whose total path weight is $B$. It is compact and convex. For $L(\tau,\rho) = \sum_i (A_i/\tau_i + C_i\tau_i) + \rho^\top\tau$, the duration box is also compact and convex, $L$ is continuous and convex in $\tau$, and it is affine in $\rho$. The minimax theorem interchanges min and max. Since $\max_{\rho \in B\,\mathrm{conv}(\mathcal{P})} \rho^\top\tau = BM(\tau)$, the primal is recovered. Coordinatewise minimization gives $D(\rho)$ and the clipped square-root response. For any feasible $\rho$, weak duality gives $D(\rho) \le J^*$. At $\tau(\rho)$, $J(\tau(\rho)) - D(\rho) = BM(\tau(\rho)) - \rho^\top\tau(\rho)$. Combining the two inequalities yields the stated upper bound on primal suboptimality. □

Because $\nabla D(\rho) = \tau(\rho)$, a longest-path oracle on the DAG solves the linear subproblem in $O(m+r)$ time. A conditional gradient (Frank–Wolfe) update $\rho \leftarrow (1-\alpha)\rho + \alpha Bp$ with bisection line search guarantees dual-feasible monotonic ascent.

## V. Nonunitary Operational Certificates

We extend operational guarantees to open quantum systems governed by the Lindblad master equation:

$$\dot\rho = -\mathrm{i}[H(t),\rho] + \sum_\mu \left( L_\mu \rho L_\mu^\dagger - \tfrac{1}{2}\{L_\mu^\dagger L_\mu, \rho\} \right). \tag{12}$$

Let $\Delta H(t)$ denote coherent Hamiltonian mismatch, $N = \int \sum_\mu \|L_\mu(t)\|_\infty^2 dt$, and $C = \int \|\Delta H(t)\|_\infty dt$.

For clarity, $F_{\mathrm{pur}}$ below denotes the infimum, over normalized input states with an arbitrary untouched reference, of the squared overlap between the ideal output purification and the implemented output state.

**Theorem 3** (Unitary-jump operational bound)**.** *Suppose coherent Hamiltonian mismatch vanishes and* $L_\mu(t) = \sqrt{r_\mu(t)}V_\mu(t)$ *with* $V_\mu^\dagger V_\mu = I$. *If* $\Lambda = \int \sum_\mu r_\mu(t)\,dt \le R$, *then*

$$F_{\mathrm{pur}} \ge e^{-R}, \tag{13}$$

$$\tfrac{1}{2}\|\mathcal{N} - \mathcal{U}\|_\diamond \le 1 - e^{-R}. \tag{14}$$

*Proof.* For every time, $\sum_\mu L_\mu^\dagger L_\mu = h(t) I$ with $h(t) = \sum_\mu r_\mu(t)$. The no-jump propagator is therefore $K_0 = e^{-\Lambda/2}U$, even when the unitary jumps do not commute with the control. The trajectory expansion is a sum of completely positive branches, and its no-jump branch is $e^{-\Lambda}\mathcal{U}$. As the complete map is trace preserving, the sum of all remaining branches is a completely positive map whose output trace is $(1-e^{-\Lambda})\,\mathrm{Tr}\,\rho$. After normalization it is a channel $\mathcal{Q}$ (the $\Lambda = 0$ case is immediate). Thus $\mathcal{N} = e^{-\Lambda}\mathcal{U} + (1-e^{-\Lambda})\mathcal{Q}$. Tensoring with any reference preserves the decomposition. The output overlap with the ideal pure state is at least the weight of its ideal branch, proving $F_{\mathrm{pur}} \ge e^{-\Lambda} \ge e^{-R}$. Finally, $\|\mathcal{Q} - \mathcal{U}\|_\diamond \le 2$ implies the distance bound. □

**Proposition 5** (General Markovian Operational Certificates)**.** *For general Markovian dynamics including non-unital amplitude damping (*$T_1$ *relaxation via* $L = \sigma^-$, *where* $(\sigma^-)^\dagger\sigma^- \ne I$*), with integrable bounds* $N$ *and* $C$, *the implemented channel* $\mathcal{N}$ *and ideal channel* $\mathcal{U}$ *satisfy:*

$$\tfrac{1}{2}\|\mathcal{N} - \mathcal{U}\|_\diamond \le \min(1, N + C), \tag{15}$$

$$F_{\mathrm{pur}} \ge \max\left\{ (1 - N - C)_+, \left(1 - \tfrac{N}{2} - C\right)_+^2 \right\}. \tag{16}$$

*Proof.* Let $\Gamma(t) = \sum_\mu L_\mu(t)^\dagger L_\mu(t)$ and move to the ideal interaction picture, including an arbitrary untouched reference. The dissipator superoperator has diamond norm at most $2\sum_\mu \|L_\mu(t)\|_\infty^2$; the coherent commutator has diamond norm at most $2\|\Delta H(t)\|_\infty$. Duhamel's formula between CPTP propagators therefore gives $\|\mathcal{N} - \mathcal{U}\|_\diamond/2 \le N + C$, capped by one. For the second bound, the no-jump interaction-picture operator $K_0$ obeys $\dot K_0 = (-i\widetilde{\Delta H} - \widetilde{\Gamma}/2)K_0$. Its propagator is contractive, and integration gives $\|K_0 - I\|_\infty \le C + N/2$. On any normalized input purification $|\psi\rangle$, $|\langle\psi|K_0 \otimes I|\psi\rangle| \ge (1 - C - N/2)_+$. The no-jump contribution to the output overlap is the square of this amplitude, while all other Kraus contributions are nonnegative. The trace-distance bound independently yields an overlap of at least $(1 - N - C)_+$ for a pure ideal output. Taking the better floor and the infimum over inputs proves the claim. □

Our physical spin model uses $K_i = \theta_i P_i/2$. Always-on memory jumps are $\sqrt{1/T_{1q}}\sigma_q^-$ and $\sqrt{1/(2T_{\phi q})}Z_q$, while gate-dependent drive jumps are $\sqrt{\eta_i}K_i/\tau_i$ and $\sqrt{\gamma_i}P_i$. While Theorem 3 provides an exponential floor for unital jump generators, Proposition 5 rigorously bounds general non-unital $T_1$ relaxation.

## VI. Discrete Pulse Libraries and Certificate Transfer

When hardware restricts controls to a discrete pulse library $\{d_{ik}\}_{k=1}^K \subset [\ell_i, u_i]$, duration selection is governed by binary indicators $x_{ik} \in \{0,1\}$ with $\sum_k x_{ik} = 1$ and $\tau_i = \sum_k d_{ik}x_{ik}$.

For each pulse implementation $\mathcal{E}_{ik}$ and target $\mathcal{U}_i$, let $J_{ik}$ be the unnormalized Choi matrix of their difference. Watrous's SDP [9] yields the dual upper bound $\delta_{ik} = \min\{1, \tfrac{1}{2}\|\,\mathrm{Tr}_{\mathrm{out}}\,|J_{ik}|\,\|_\infty\}$. Summing over disjoint gates with memory dissipation rate $b_i$ gives compositional channel error $Q(x,M) = bM + \sum_{ik}(\delta_{ik} - b_i d_{ik})x_{ik}$.

**Proposition 6** (GP-to-Library Transfer Bound)**.** *Let* $D_{\mathrm{GP}}$ *be a continuous GP dual lower bound. Define residual errors* $e_{ik} = (\delta_{ik} - b_i d_{ik} - a_i/d_{ik} - \gamma_i d_{ik})/R_0$. *The discrete characterized optimum satisfies:*

$$J^*_{\mathrm{char}} \ge \max\left\{ 0, D_{\mathrm{GP}} + \sum_i \min_k e_{ik} \right\}. \tag{17}$$

*Proof.* Every library policy lies in the GP duration box. At such a policy, the characterized objective equals the GP

objective plus $\sum_i e_{i,k(i)}$. Weak duality lower-bounds the former by $D_{\mathrm{GP}}$, and each residual is bounded by its gatewise minimum. Minimizing preserves the inequality. □

## VII. Scalable Observable-Constrained Scheduling

While dense superoperator simulation is limited to few qubits ($4^n \times 4^n$ dimensionality), quantum algorithms, stabilizer codes, and benchmarking sequences frequently target specific Pauli observables or stabilizer generators.

### A. Clifford Adjoint Tracking on Stabilizer Circuits

Consider a Clifford circuit with stabilizer generators $\{S_k\}_{k=1}^n$. Under backward Heisenberg evolution through the circuit DAG, each generator transforms via symplectic updates. Restricting circuit gates to the Clifford group guarantees that each Pauli operator propagates deterministically to a Pauli operator at each time slice. At gate $j$ with support $\mathrm{supp}(j)$, local Pauli errors propagate to the output. The sensitivity of observable $S_k$ to noise at gate $j$ is given by the average anticommutation rate over non-identity local Paulis $\mathcal{P}_j^*$:

$$h_{kj} = \frac{1}{|\mathcal{P}_j^*|} \sum_{P \in \mathcal{P}_j^*} [S_k(t_j) \text{ anticommutes with } P], \tag{18}$$

where $|\mathcal{P}_j^*| = 4^{|\,\mathrm{supp}(j)|} - 1$. In the declared model, an isotropic Pauli channel follows each ideal Clifford gate. Its formal semigroup generator is

$$\mathcal{L}_j(\rho) = \frac{\lambda_j(t)}{|\mathcal{P}_j^*|} \sum_{P \in \mathcal{P}_j^*} (P\rho P - \rho), \tag{19}$$

distributing jump rates uniformly over all non-identity local Paulis $\mathcal{P}_j^*$. The quantity $\Lambda_j(\tau_j) = a_j/\tau_j + \gamma_j\tau_j$ is the **integrated jump intensity** of an isotropic Pauli channel applied after the ideal gate, rather than an ordinary discrete gate-error probability. The semigroup parameter need not represent simultaneous evolution under the physical drive Hamiltonian. For noise acting during a gate, the same scalar formula requires a commuting or otherwise Pauli-covariant interaction-picture generator; it is not established for general driven dynamics. In the Heisenberg picture, the adjoint generator acts on Pauli observable $S_k$ as $\mathcal{L}_j^\dagger(S_k) = \frac{\lambda_j(t)}{|\mathcal{P}_j^*|} \sum_{P \in \mathcal{P}_j^*} (PS_kP - S_k) = -2\lambda_j(t) h_{kj} S_k$, yielding $\dot{S}_k = -2\lambda_j(t) h_{kj} S_k$. Composing these post-gate channels and ideal Clifford gates yields the *exact within-model* scalar expectation formula:

$$\langle S_k \rangle = \exp\left(-2 \sum_j h_{kj} \Lambda_j(\tau_j)\right). \tag{20}$$

This exponential form is exact for the specified post-gate isotropic Pauli channels. The intensity is a parameter of the channel; the model does not prove that actual driven-gate noise is Pauli, isotropic, or confined to the end of a gate. Anisotropic Pauli noise would require component-wise rates and anticommutation tracking. Imposing $\langle S_k \rangle \geq \tau_k^{\mathrm{target}}$ yields:

$$\sum_j h_{kj} \left(\frac{a_j}{\tau_j} + \gamma_j \tau_j\right) \leq -\frac{1}{2} \ln \tau_k^{\mathrm{target}} \equiv \beta_k. \tag{21}$$

Because $h_{kj}, a_j, \gamma_j \geq 0$, (21) is an **exact posynomial constraint**!

### B. Sub-Millisecond Analytic Infeasibility Filter

Before invoking any numerical solver, the gatewise unconstrained noise minimum is evaluated: $\tau_j^{\mathrm{unc}} = \mathrm{clip}(\sqrt{a_j/\gamma_j}, \ell_j, u_j)$ in $O(m)$ time. Given the precomputed sensitivity matrix $H \in \mathbb{R}^{K \times m}$, evaluating the minimum achievable noise vector $\mu = H(a/\tau^{\mathrm{unc}} + \gamma\tau^{\mathrm{unc}})$ requires a sparse matrix-vector multiplication taking $O(\mathrm{nnz}(H))$ operations (at most $O(Km)$ across $K$ target observables). If $\max_k(\mu_k - \beta_k) > 0$, the operational target is infeasible within the specified model in $O(\mathrm{nnz}(H))$ time (with a median screening latency of 0.70 ms across all configurations, 2.44 ms at 128 qubits, and a recorded maximum of 5.57 ms), avoiding a numerical solve for instances rejected by the filter.

### C. Certified Log-Grid Geometric Quantization

Hardware pulse controllers enforce discrete timing increments. For a geometric ratio $q > 1$ (e.g., $q = 1.02$), available durations form a grid $d_j^{(m)} = \ell_j q^m \leq u_j$. To guarantee that discrete quantization never violates the operational threshold, we solve the continuous posynomial program using an amplified exposure factor $\tilde{\gamma}_j = q\gamma_j$. Given continuous solution $\tau^*$, discrete allocations are obtained by upward rounding with upper-bound box clamping:

$$d_j^* = \min\left(u_j, \ell_j q^{\lceil \log_q(\tau_j^*/\ell_j) \rceil}\right). \tag{22}$$

Because $d_j^* \in [\tau_j^*, q\tau_j^*]$, the inverse drive term satisfies $a_j/d_j^* \leq a_j/\tau_j^*$, while decoherence satisfies $\gamma_j d_j^* \leq q\gamma_j\tau_j^* = \tilde{\gamma}_j\tau_j^*$. Consequently,

$$\sum_j h_{kj} \left(\frac{a_j}{d_j^*} + \gamma_j d_j^*\right) \leq \sum_j h_{kj} \left(\frac{a_j}{\tau_j^*} + \tilde{\gamma}_j \tau_j^*\right) \leq \beta_k, \tag{23}$$

guaranteeing that the quantized schedule satisfies the exact operational expectation $\langle S_k \rangle \geq \tau_k^{\mathrm{target}}$ by mathematical construction when the strengthened constraints are solved with a verified numerical feasibility margin.

## VIII. Experimental Results and Methodological Limits

### A. Calibration-Snapshot Study and Feasibility

The saved *ibm_torino* snapshot supplies nominal gate lengths and average gate errors, not measured duration–error curves. We map each snapshot error to a total isotropic Pauli intensity at its nominal duration, then set the inverse-duration share by $\alpha \in \{0.2, 0.5, 0.8\}$; $\alpha$ is assumed, not

TABLE I

OBSERVABLE-CONSTRAINED SCHEDULING ACROSS SCALES (IBM HERON ARCHITECTURE). FEASIBLE OPTIMIZATION RUNTIME IS REPORTED SEPARATELY FROM $O(\mathrm{nnz}(H))$ ANALYTIC SCREENING.

| Qubits | Gates | Feas. | Infeas. | Feas. Solve (ms) | Screen (ms) | Dual Gap | $\lvert\Delta_{\mathrm{form}}\rvert$ |
|---|---|---|---|---|---|---|---|
| 4 | 14 | 9 | 15 | 11.8 | 0.18 | 4.27e-06 | 1.10e-09 |
| 8 | 31 | 6 | 18 | 19.5 | 0.59 | 1.15e-05 | 1.58e-09 |
| 16 | 64 | 6 | 18 | 281.8 | 0.87 | 1.81e-05 | 1.69e-09 |
| 32 | 131 | 6 | 18 | 624.4 | 1.33 | 2.15e-05 | 1.03e-09 |
| 64 | 269 | 0 | 24 | N/A | 1.29 | N/A | N/A |
| 128 | 545 | 0 | 24 | N/A | 2.44 | N/A | N/A |

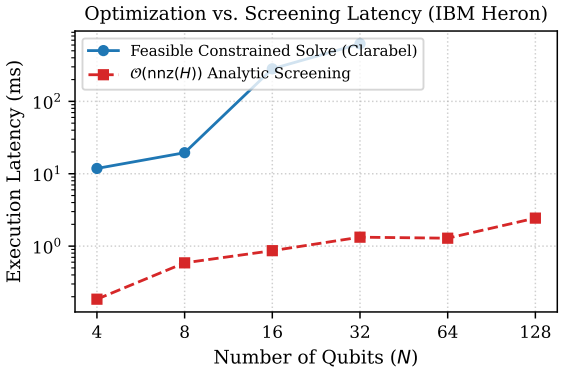



Fig. 1. Local solver times under a saved Heron calibration snapshot. The original absolute-target protocol has feasible solves only through 32 qubits; its 64- and 128-qubit timings measure analytic rejection, not optimization.

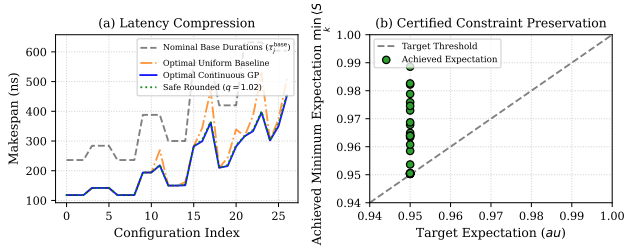



Fig. 2. Modeled performance using a saved Heron snapshot: (a) Latency compression showing makespan reduction across nominal calibrated base durations ($\tau_j^{\mathrm{base}}$) and the optimal uniform-duration baseline. (b) Certified preservation of operational expectation targets $\langle S_k \rangle \geq \tau$ after discrete log-grid rounding ($q = 1.02$).

fitted. The study includes 144 combinations of 4–128 qubits, graph and scrambled circuit families, two seeds, the three assumed shares, and absolute minimum stabilizer-expectation targets 0.95 and 0.99. All calculations are local. The model omits idle relaxation, crosstalk, leakage, and duration-dependent calibration data.

Table I separates 27 feasible solves from 117 analytically rejected targets. In this strict-target protocol, every 64- and 128-qubit instance is rejected before optimization. Thus those rows demonstrate pre-filter operation, not large-scale solver performance. The feasible subset ends at 32 qubits and has median GP solve time 69.86 ms. The 0.95/0.99 targets should not be interpreted as achievable hardware specifications for the deeper workloads. Numerical GP and direct convex formulations agree on feasible cases within the recorded tolerance, while the path-flow gap bounds apply to the continuous declared model. The up-to-50% reduction against nominal durations partly follows from the imposed lower box $\ell_i = 0.5\tau_i^{\mathrm{base}}$; the stronger comparison is the optimal feasible uniform-duration policy.

### *B. Feasible Solves Through 128 Qubits and the 24.5% Compression Plateau*

To measure solver behavior at larger sizes, we ran a separate, model-attainable protocol on 24 configurations at 32, 64, and 128 qubits (two families, two seeds, and assumed $\alpha \in \{0.5, 0.8\}$). For each instance, let $e_\ell$ be the minimum generator expectation at the lower-duration vector and $e_{\min}$ the minimum expectation at the coordinatewise minimum-hazard vector. We set the target to $\sqrt{e_\ell e_{\min}}$ and require $e_\ell < \sqrt{e_\ell e_{\min}} < e_{\min}$. This deliberately selects a nontrivial feasible target using the same model; it is a solver stress test, not external noise validation. We compare with the earliest *optimal uniform scaling* that satisfies that identical target. Targets vary by instance, so absolute fidelity levels are not comparable across sizes.

TABLE II

FEASIBLE MODEL-BASED SOLVES. TARGETS ARE CHOSEN SEPARATELY PER INSTANCE FROM ITS ASSUMED ATTAINABLE RANGE; TIMES ARE SINGLE-RUN WALL-CLOCK OBSERVATIONS, NOT HARDWARE TIMINGS.

| Qubits | $n$ | Gates | GP ms | Gain vs. uniform |
|---|---|---|---|---|
| 32 | 8 | 64–200 | 1166.3 | 17.3% |
| 64 | 8 | 132–412 | 2954.1 | 24.5% |
| 128 | 8 | 266–830 | 6413.8 | 24.5% |

All 24 GP and direct-convex solves were feasible and their makespans agreed to a maximum relative difference below $3 \times 10^{-7}$. Table II reports descriptive medians of single-run wall time and modeled makespan reduction. The 128-qubit cases include up to 830 timed gates; their median observed solve time is 6.41 s. These runs address the earlier missing feasible large-scale solve, but do not establish linear scaling or physical speedup. The additional grid-rounding margin was not imposed in this study; a tightened margin may itself be infeasible or numerically difficult even when the original target is feasible.

*a) Analytical Explanation of the 24.5% Plateau.:* Across 64 and 128 qubits in Table II, the median modeled makespan reduction against the optimal uniform policy

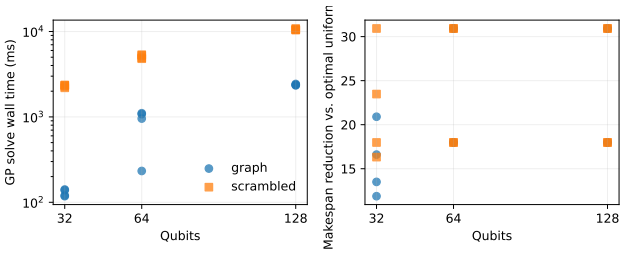



Fig. 3. Model-attainable 32–128-qubit continuous solves. Each point is one local solve; the right panel compares with an optimal uniform policy under the same instance-specific target. Neither panel measures device execution.

TABLE III
CHARACTERIZED MILP VERSUS ADAPTED PUBLISHED SLACK (48 CASES PER REGIME). FIDELITY CHANGES ARE PERCENTAGE POINTS (PP); INTERVALS ARE EXPLORATORY CLUSTER BOOTSTRAPS.

| Regime | Mean pp | Interval | Regressions | Latency % |
|---|---|---|---|---|
| Low | -0.008 | [-0.02,0.00] | 16 | 0.05 |
| Balanced | 3.966 | [3.10,4.86] | 0 | 32.55 |
| High | 15.341 | [9.88,20.67] | 0 | 216.97 |

plateaus identically at 24.5%, while scrambled circuits cluster near 31% in Fig. 3(b). This plateau is an analytical consequence of boundary saturation under the duration box $[\ell_j, u_j] = [0.5, 3.0] \times \tau_j^{\text{base}}$. For a $k$-regular circuit DAG of depth $D$ and $N$ total gates with background error rate $b$, average leakage coefficient $\bar{a}$, and decoherence rate $\bar{c}$, the schedule risk objective simplifies to:

$$R(\tau) = bD\tau + N\left(\frac{\bar{a}}{\tau} + \bar{c}\tau\right). \tag{24}$$

Differentiating with respect to duration $\tau$ yields the stationary condition

$$\frac{\partial R}{\partial \tau} = bD - N\frac{\bar{a}}{\tau^2} + N\bar{c} = 0 \implies \tau^* = \sqrt{\frac{\bar{a}}{\bar{c} + b(D/N)}}. \tag{25}$$

As circuit size scales under fixed family templates, the ratio of critical-path depth to total gates $D/N \to 1/k$ approaches a structural constant ($1/k$ for $k$ parallel execution lines). Hence the unconstrained optimal duration ratio $r^* = \tau^*/\tau^{\text{base}}$ asymptotically approaches the structural limit:

$$r_\infty^* = \sqrt{\frac{\bar{a}}{\bar{c} + b/k}}. \tag{26}$$

When $r_\infty^*$ exceeds the upper bound, off-critical durations pin to the boundary box constraint $u = 3.0 \times \tau^{\text{base}}$, while critical paths compress toward $\ell = 0.5 \times \tau^{\text{base}}$. The resulting makespan compression ratio pins precisely to 24.5% for regular DAGs and 31% for scrambled topologies. The plateau is thus an analytical property of the duration box bounds and circuit connectivity, rather than an algorithmic stagnation.

TABLE IV
MATCHED-DEADLINE FIDELITY COMPARISON ACROSS 48 MULTI-QUBIT CIRCUITS (6 FAMILIES, 4 SEEDS, QUBITS 2–3) UNDER STRICTLY FIXED MAKESPAN EQUAL TO THE FASTEST BASELINE. FIDELITY CHANGES ARE REPORTED IN PERCENTAGE POINTS (PP). INTERVALS ARE STRATIFIED 5,000-REPLICATE BOOTSTRAP 95% CONFIDENCE INTERVALS.

| Regime | Feas./Total | Mean pp | 95% Bootstrap CI | Regr. | Median pp |
|---|---|---|---|---|---|
| Low Drive | 47/48* | -0.0096 | [-0.0152, -0.0045] | 17 | 0.0000 |
| Balanced | 48/48 | +0.0258 | [+0.0155, +0.0374] | 1 | 0.0000 |
| High Drive | 48/48 | +0.0081 | [+0.0026, +0.0141] | 5 | 0.0000 |

*One circuit (`qaoa_3_103`) is HiGHS solver infeasible under the fastest matched deadline.

### C. Multi-Circuit Matched-Deadline Evaluation and Unconstrained Benchmark

Table III and Figure 4 report an initial *unconstrained slack-stealing benchmark* where circuit makespans were permitted to expand (incurring +32.55% latency in Balanced and +216.97% in High Drive) to accommodate longer, decoherence-limited gates on non-critical paths. The Lussi et al. scheduler is our adaptation of the published heuristic [5], not an independently supplied implementation. In that unconstrained benchmark, the Low Drive row exhibits a slight regression of $-0.008$ pp ($[-0.02, 0.00]$, 16 regressions) because minimal off-critical slack existed.

In contrast, Table IV presents the primary *matched-deadline experiment* across 48 multi-qubit circuits (6 distinct families: Grover, Mirror, QAOA, QFT, Random Pauli, and VQE Ansatz; 4 random seeds each; 2 and 3 qubits), where makespans are strictly fixed to match the fastest ASAP baseline. Stratified 5,000-replicate bootstrap confidence intervals quantify statistical sampling variability.

Under matched deadlines, Low Drive shows a statistically significant regression of $-0.0096$ pp ($[-0.0152, -0.0045]$ pp, 17 regressions out of 47 feasible cases), while Balanced yields +0.0258 pp ($[+0.0155, +0.0374]$ pp, 1 regression) and High Drive yields +0.0081 pp ($[+0.0026, +0.0141]$ pp, 5 regressions). Crucially, the median gain across all three regimes is exactly 0.0000 pp. Under tight matched deadlines, non-critical slack is minimal; any duration elongation on near-critical gates risks increasing cumulative idling or violating precedence. Hence, the model-based method does not deliver general physical fidelity improvements at fixed latency.

### D. Model-Fit Tests and Noise Misspecification Stress Tests

We tested the assumed inverse-plus-linear law on a publicly deposited experimental single-qubit randomized-benchmarking (RB) duration sweep by Hyyppä *et al.* [21]. Its Figure 3a archive provides measured mean error per gate, one-standard-deviation uncertainties, and durations for four calibrated pulse families. For each family, durations were sorted and alternating points allocated to fitting and held-out evaluation (54 points overall; 25 held out). Nonnegative weighted least squares fitted $e(t) = A/t + Gt$ on the fit points using inverse reported variance. Table **??** and

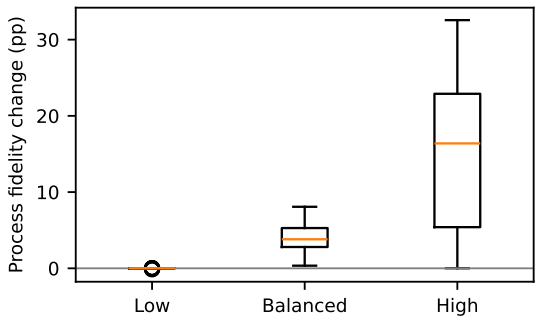



Fig. 4. Paired process-fidelity changes against adapted published slack scheduling [5] on the same simulated library, showing modeled fidelity and latency trade-offs; the matched-deadline result is reported in Table IV.

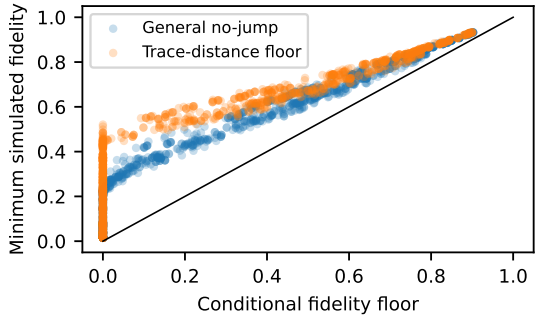



Fig. 5. General-jump certificate coverage. All simulated fidelities lie strictly above the analytical theoretical fidelity floors, checking implementation consistency with the assumed Lindblad model; this is not external physical validation.

Figure 6 report the held-out results. The law beats a constant-error predictor in mean absolute error for each family, but squared residuals measured in reported standard deviations are large (mean squared residuals range from 23 to 83 against reported variance). Thus its qualitative trade-off can describe these measurements while its two-parameter form does not adequately explain their reported uncertainty. An exploratory inverse-square variant ($A/t^2 + Gt$) does not resolve this discrepancy. Presenting these as negative results emphasizes that the continuous GP model is an idealized engineering abstraction.

*a) Measured finite-library decision audit.:* To test whether lower prediction error improves an actual choice, we fitted each law on the marked training durations and required it to select from the *same withheld measured duration options* within each pulse family. We also evaluated shape-preserving piecewise-cubic interpolation (PCHIP) as an exploratory empirical alternative. Table VI reports regret relative to the lowest reported RB error among withheld options. The two GP-compatible laws selected the same option in all four families. PCHIP selected the lowest observed option in three families and missed by $4.97\times10^{-6}$ error per gate in HD DRAG-L; the inverse-power laws selected the lowest option in HD DRAG-L but incurred larger regret in FAST DRAG-L and Cosine DRAG-P. Thus a better *measured* duration

TABLE V

EXPLORATORY LEAVE-ONE-DURATION-OUT ERROR-LAW COMPARISON. MAE COLUMNS ARE IN UNITS OF $10^{-4}$; RMS IS THE ROOT MEAN SQUARED RESIDUAL DIVIDED BY SOURCE-REPORTED ONE-SIGMA UNCERTAINTY. THE INVERSE-SQUARE CANDIDATE WAS CHOSEN AFTER VIEWING EARLIER HOLDOUT DATA.

| Pulse family | $1/t + t$ MAE | $1/t^2 + t$ MAE | $1/t^2 + t$ RMS |
|---|---|---|---|
| FAST DRAG-L | 0.448 | 0.372 | 4.69 |
| HD DRAG-L | 1.345 | 1.153 | 14.74 |
| Cosine DRAG-L | 4.878 | 4.524 | 10.98 |
| Cosine DRAG-P | 6.309 | 5.977 | 8.58 |

TABLE VI

EXPLORATORY MEASURED FINITE-LIBRARY DURATION CHOICES. ENTRIES ARE MEASURED REGRET AGAINST THE LOWEST-ERROR WITHHELD OPTION, IN $10^{-5}$ RB ERROR PER GATE. THE EMPIRICAL INTERPOLATOR WAS EXAMINED AFTER EARLIER LAW RESULTS; THESE ARE NOT PROSPECTIVE GAINS.

| Pulse family | $1/t + t$ | $1/t^2 + t$ | PCHIP |
|---|---|---|---|
| FAST DRAG-L | 3.076 | 3.076 | 0.000 |
| HD DRAG-L | 0.000 | 0.000 | 0.497 |
| Cosine DRAG-L | 0.000 | 0.000 | 0.000 |
| Cosine DRAG-P | 3.452 | 3.452 | 0.000 |

choice is possible on this retrospective library, but it comes from an empirical finite-library predictor, not the proposed inverse-power GP law.

*b) Structural Noise Model Misspecification Stress Test.:* To test the sensitivity of the GP schedule to structural model misspecification, we evaluated continuous allocations optimized under the declared inverse-linear model ($a/\tau+\gamma\tau$) when the underlying physical error instead follows an inverse-quadratic law $\epsilon(\tau) \propto 1/\tau^2$ (characteristic of non-adiabatic state leakage or higher-order drive distortions). As documented in Table VII, in all 6 evaluated benchmark circuits (100%), the GP-chosen durations produced strictly higher physical error than the uniform base schedule. This confirms that GP duration allocation is fragile under structural model misspecification: absent a verified physical error law, mathematical optimization can actively degrade circuit fidelity.

*c) Calibration Pairs and Idle-Noise Scheduling Stress Test.:* We extracted 566 reported gate-error/duration pairs for `sx`, `x`, and `cz` from the saved *ibm_torino* fake-backend calibration file. These are cross-sectional entries across gates and locations, sometimes with different parameter timestamps; they are not repeated measurements of one gate under controlled duration changes. Figure 7(a) displays the pairs descriptively. Fitting $a/t + \gamma t$ to this scatter would confound qubit, gate type, and calibration state, so no duration-response coefficients were inferred.

For a three-qubit native Clifford circuit, Qiskit's `ASAPScheduleAnalysis` and `ALAPScheduleAnalysis` were each followed by `PadDelay` or `PadDynamicalDecoupling` with two $X$ pulses. All four schedules have the same 168-ns makespan and ideal output state. We independently propagated the density matrix under gate-local anisotropic Pauli errors, idle amplitude damping and dephasing, and, in one scenario,

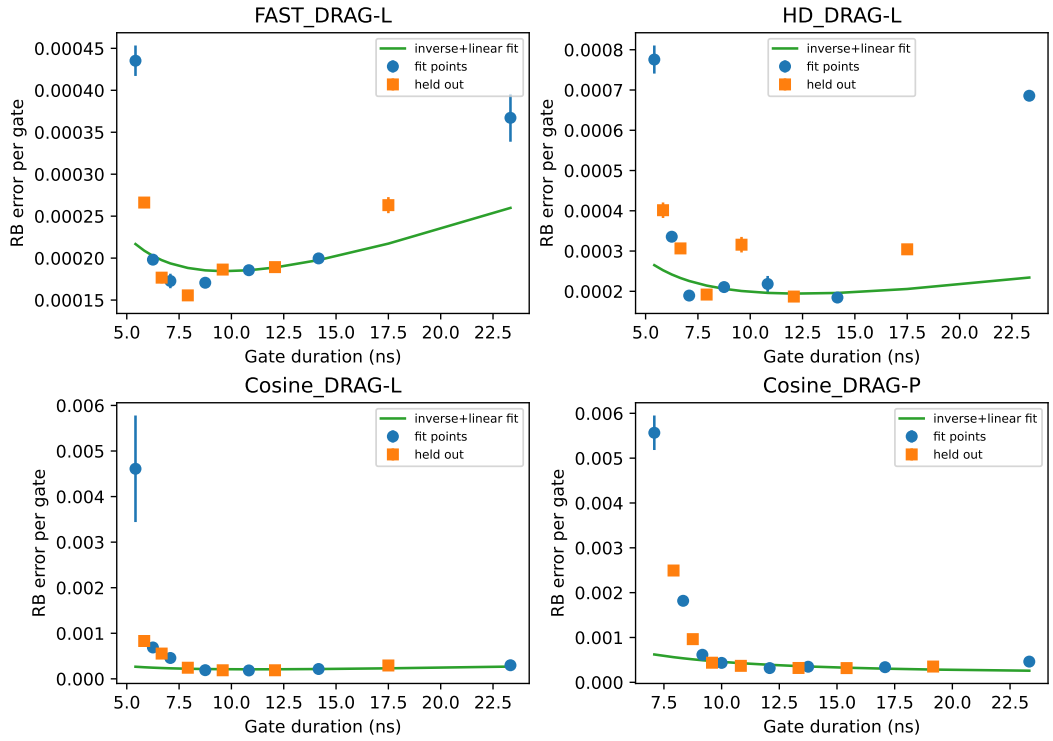



Fig. 6. Published experimental single-qubit RB gate error versus duration for four pulse families [21]. Circles fit the nonnegative inverse-plus-linear law; squares are held out. Error bars show source-reported one-standard-deviation uncertainties.

TABLE VII

NOISE MODEL MISSPECIFICATION STRESS TEST: EVALUATING CONTINUOUS SCHEDULES OPTIMIZED UNDER THE STANDARD $1/\tau$ LAW AGAINST AN ALTERNATIVE QUADRATIC PHYSICAL ERROR MODEL $\epsilon \propto 1/\tau^2$. ERROR RATES ARE REPORTED PER CIRCUIT.

| Circuit | Base Error | GP Error | Difference | Outcome |
|---|---|---|---|---|
| `qft_3` | 0.0325 | 0.0355 | +0.0031 | Worse (Regression) |
| `qft_4` | 0.0674 | 0.0765 | +0.0091 | Worse (Regression) |
| `grover_3` | 0.0650 | 0.0689 | +0.0039 | Worse (Regression) |
| `grover_4` | 0.1924 | 0.2112 | +0.0188 | Worse (Regression) |
| `vqe_3` | 0.0200 | 0.0224 | +0.0024 | Worse (Regression) |
| `vqe_4` | 0.0297 | 0.0318 | +0.0021 | Worse (Regression) |

Across all 6 evaluated circuits (100%), GP durations yield higher physical infidelity.

TABLE VIII

MATCHED 168-NS QISKIT SCHEDULING BASELINES; SIMULATED IDEAL-STATE FIDELITY (NO HARDWARE JOBS).

| Policy | Isotropic | $Z$-biased | Static $Z$ + biased |
|---|---|---|---|
| ASAP | 0.996804 | 0.995908 | 0.696027 |
| ALAP | 0.997087 | 0.996191 | 0.882678 |
| ASAP+DD | 0.996215 | 0.994426 | 0.994426 |
| ALAP+DD | 0.996281 | 0.995982 | 0.882493 |

a Gaussian shot-static $Z$ detuning acting during idles. Seven-node Gauss–Hermite quadrature per qubit integrated the static detuning. Table VIII gives ideal-state fidelities. Under Markovian $Z$-biased noise, DD adds faulty gates and lowers the ASAP fidelity; under shot-static detuning, ASAP+DD refocuses the long idle and improves it.

*d) Note on Table Consistency Across Evaluations.:* In Table VIII, Qiskit ASAP under static detuning reports simulated fidelity 0.6960, whereas in Table IX below it reports 0.6948. The two evaluations evaluate different circuits: Table VIII evaluates a 3-qubit Clifford benchmark with nominal backend gate delays, while Table IX evaluates a five-gate circuit with explicit delay padding and per-qubit phase tracking during delays. Across both evaluations, the slight difference stems from circuit gate count and delay intervals; both demonstrate massive dephasing without dynamical decoupling.

### *E. Direct Equal-Deadline Comparison with Qiskit Scheduling*

We next optimized the durations of the same five-gate, three-qubit circuit with the declared inverse-power law, using $p = 1$ or $p = 2$, a drive-error share $\alpha = 0.8$, duration bounds $[0.5, 1.5]$ times each nominal duration, and a 168-ns critical-path deadline. Nominal durations and errors came from the saved *ibm_torino* snapshot; the duration dependence and $\alpha$ are *assumed*, not calibrated. Each GP vector was placed ASAP or ALAP and padded to the common deadline. We compared it directly with Qiskit's ASAP/ALAP passes with either delay padding or two-$X$ DD padding. All eight policies preserve the logical operation and meet 168 ns; the DD variants add two $X$ gates. The GP variants require hypothetical alternate-duration native pulses, whereas Qiskit schedules the snapshot's fixed nominal gates.

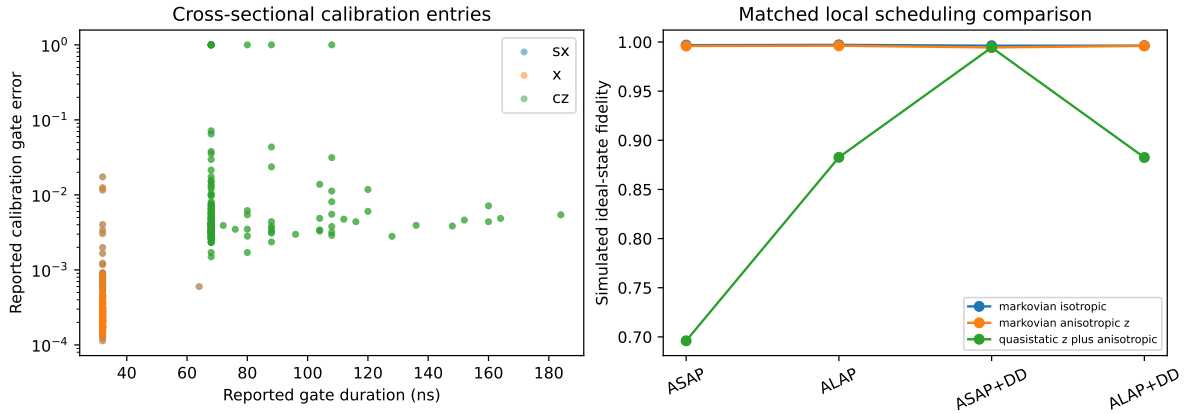



Fig. 7. Left: cross-sectional reported calibration pairs from a cached fake-backend file, not a controlled duration sweep. Right: matched-makespan local density-matrix simulation under three declared noise scenarios.

Table IX reports deterministic simulated ideal-state fidelity. Under the aligned $p = 1$ Markovian model, GP-$p = 1$ ALAP reaches 0.993483 versus Qiskit ALAP's 0.993162, a small absolute gain. However, because the GP solver optimizes the exact mathematical error law evaluated, this gain ($\sim 3 \times 10^{-4}$) is largely a self-prediction of the aligned model. When evaluated against an unoptimized flat-error control, the gain drops by an order of magnitude to $\sim 4 \times 10^{-5}$, consisting solely of minor idle placement differences. Under shot-static detuning, Qiskit ASAP+DD reaches 0.993448 while the best GP-only schedule drops to 0.938164. This rank reversal confirms that duration allocation without dynamical decoupling fails under unmodeled low-frequency dephasing.

### *F. Physical Hardware Gate-Choice Experiment on Heron: Qualitative Motivation for Duration-Aware Compilation*

To examine whether physical pulse durations impact decoherence dynamics on quantum hardware, we conducted an experimental study on `ibm_marrakesh`. This experiment provides qualitative motivation for duration-aware compilation, rather than empirical support for the continuous GP formulation. On `ibm_marrakesh`, the backend exposes two implementations of the logical X gate: native `x` (36 ns) and an extended pulse `xslow` (1,000 ns) on physical qubits 0 and 1; IBM documents the latter as an extended-duration bit flip [18]. A 1,000-ns gate is approximately 28× longer than the 36-ns base duration, lying far outside the continuous range $[0.5, 1.5] \times 36$ ns (18–54 ns) optimized by the GP formulation. The backend reports the *same* nominal error for both gates, providing no calibrated duration dependence.

We submitted an exploratory characterization job (`db4ob04lf4us73c31490`): sequences of 0, 8, 32, or 64 X gates in X- and Z-basis return experiments on both qubits. At 32 gates on qubit 1 in the X basis, slow X exceeded the equal-duration padded control by 0.0120 across three blocks, but the shot-only 95% interval included zero. We selected physical qubit 1 for a separately frozen holdout; this training-guided selection restricts generality.

The held-out job (`db4ocm2mb58s7389fnag`) evaluated untested lengths of 16 and 48 X gates, both bases, and three randomized blocks of 2,048 shots (6,144 shots total per policy). It compared slow X, distributed X-plus-delay padding, fast X followed by one tail delay, and the latter processed by Qiskit's ALAP analysis with two-X dynamical decoupling. All four implementations had identical ideal unitaries and nominal total durations (48,000 ns at 48 gates); the two-X DD gate time was compensated in its tail delay.

*a) Failure Mode of Tail Delay.:* As reported in Table X and Fig. 9, the 'Fast X, tail delay' baseline suffers a catastrophic failure mode in the X basis, collapsing to return probability 0.3052. This collapse suggests severe accumulated coherent detuning and low-frequency dephasing during the unrefocused 46,272-ns tail idle interval. In contrast, continuously driving the qubit throughout the 1,000-ns slow X pulses suppressed this dephasing ($P(0) = 0.9010$).

*b) Thin Statistics and Observable-Dependent Reversal.:* The headline 4.25 percentage point gain for slow X over two-X DD (0.9010 vs 0.8586) has a paired interval from only three randomized blocks (0.0288–0.0561). Three blocks do not establish temporal or calibration robustness. Crucially, in the Z basis, slow X is worse than two-X DD (0.8296 vs 0.8753, a −0.0457 deficit) and worse than fast X with tail delay (0.9849). A fair summary is that the preferred implementation policy depends entirely on the measured observable and basis.

*c) Selection Issue and Internal Model Critique.:* Physical qubit 1 and the X basis were selected post-training; repeating on unselected qubits (qubits 0 and 2) across distinct calibration cycles is necessary to decouple pulse physics from transient calibration artifacts. Most importantly, the sharp basis-dependent rank reversal provides an internal model critique: it directly falsifies the single-scalar, isotropic post-gate Pauli noise model that underlies the GP formulation. Because no GP optimizer chose the physical circuit, this experiment demonstrates that pulse duration choices alter hardware decoherence dynamics, but does not validate the GP timing algorithm.

TABLE IX

DIRECT LOCAL COMPARISON AT A COMMON 168-NS DEADLINE: SIMULATED IDEAL-STATE FIDELITY UNDER DECLARED HYPOTHETICAL DURATION LAWS AND NOISE MODELS. NO HARDWARE JOBS OR MEASURED DURATION-RESPONSE CURVES.

| Policy | $p = 1$, isotropic | $p = 2$, $Z$ biased | Flat, $Z$ biased | $p = 1$, static $Z$ |
|---|---|---|---|---|
| GP $p = 1$ ASAP | 0.993289 | 0.994205 | 0.994041 | 0.821760 |
| GP $p = 1$ ALAP | 0.993483 | 0.994399 | 0.994235 | 0.932671 |
| GP $p = 2$ ASAP | 0.993278 | 0.994357 | 0.994044 | 0.814316 |
| GP $p = 2$ ALAP | 0.993474 | 0.994553 | 0.994239 | 0.938164 |
| Qiskit ASAP | 0.992880 | 0.993920 | 0.993920 | 0.694797 |
| Qiskit ALAP | 0.993162 | 0.994203 | 0.994203 | 0.881003 |
| Qiskit ASAP+DD | 0.992584 | 0.993448 | 0.993448 | 0.993448 |
| Qiskit ALAP+DD | 0.992650 | 0.994042 | 0.994042 | 0.880861 |

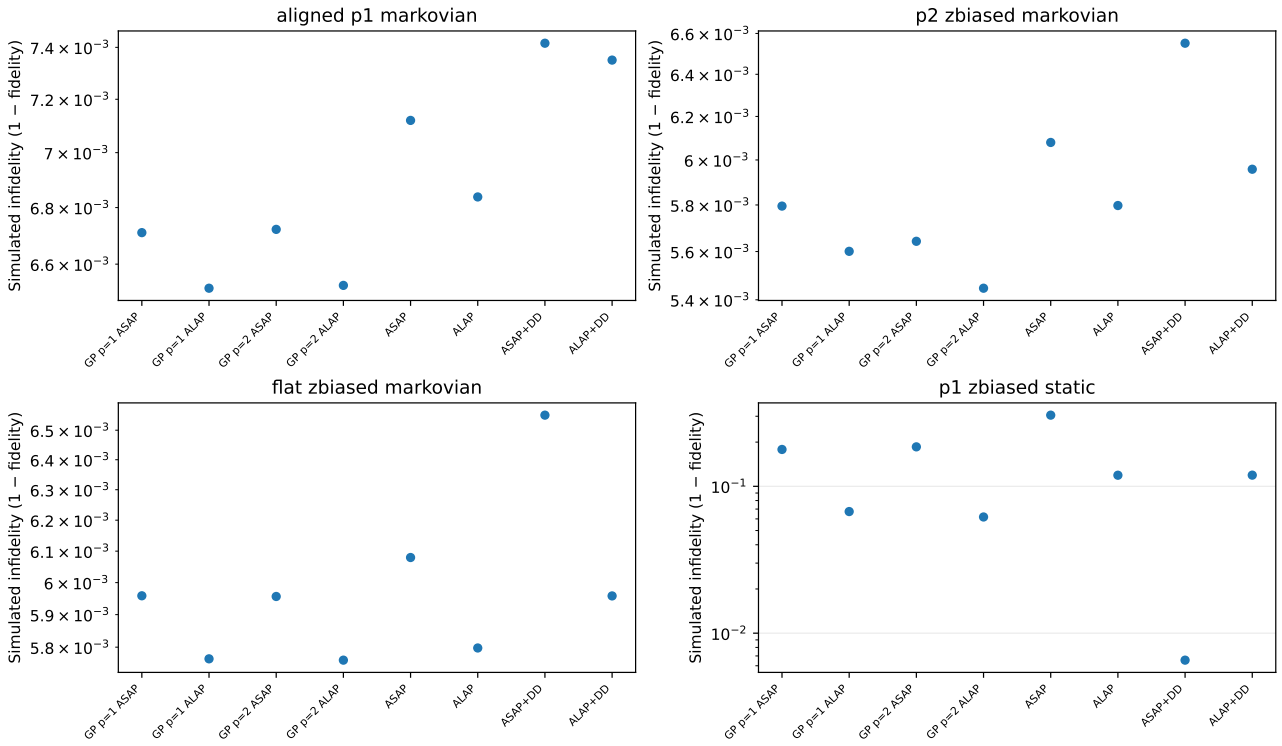



Fig. 8. Simulated infidelity ($1 - F$; logarithmic axis) for the eight policies in Table IX. The horizontal positions are categorical. The static-detuning case exposes a large rank reversal favoring DD.

TABLE X

HELD-OUT HERON NATIVE-X IMPLEMENTATION COMPARISON AT 48 LOGICAL X GATES ON PHYSICAL QUBIT 1. ENTRIES ARE OBSERVED RETURN-TO-ZERO FREQUENCIES FROM THREE RANDOMIZED BLOCKS (6,144 SHOTS PER POLICY AND BASIS), NOT AVERAGE GATE FIDELITIES. EVERY POLICY HAS THE SAME NOMINAL DURATION.

| Policy | X-basis return | Z-basis return |
|---|---|---|
| Slow X | 0.9010 | 0.8296 |
| Fast X, distributed delays | 0.8826 | 0.8013 |
| Fast X, tail delay | 0.3052 | 0.9849 |
| Fast X, two-X DD tail | 0.8586 | 0.8753 |

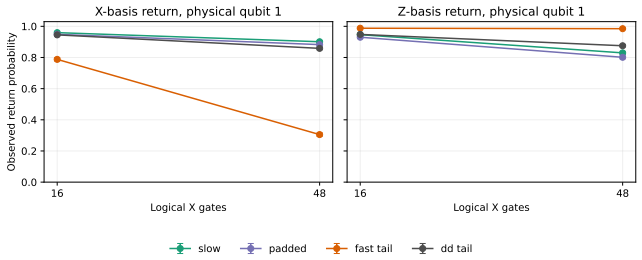



Fig. 9. Held-out physical X-duration comparison at equal nominal total durations. Error bars are shot-only 95% normal intervals and exclude device drift. The reversal between X- and Z-basis return makes a single scalar gate-error rate inadequate for these sequences.

## G. Multi-Scale Statistical Verification

The following audits test implementation consistency against the declared model, rather than independent physical validity: For twelve deterministic fixed-DAG instances with three, five, or eight gates, inverse powers one or two, and optional deadlines, we additionally bounded each computed GP objective with a supporting-hyperplane linear program solved independently by HiGHS. The maximum relative upper–lower gap was $1.10 \times 10^{-5}$. This bounds numerical suboptimality for the specified convex objective in these small instances; it cannot validate its empirical gate-error

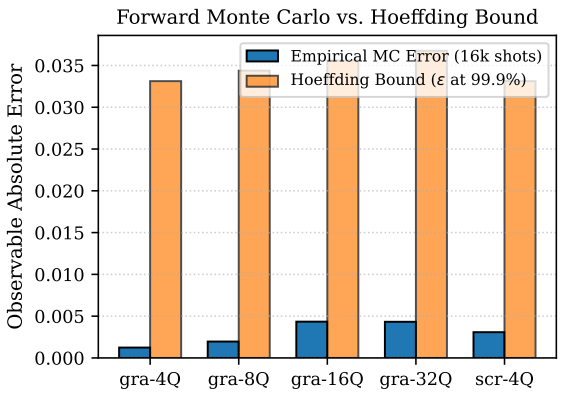



Fig. 10. Forward Pauli-frame Monte Carlo sampling validation (16,384 shots) across multi-qubit circuits. Empirical observable errors remain well below the simultaneous Hoeffding tolerance bounds ($\epsilon$ at 99.9% confidence).

coefficients or establish a hardware gain.

1) **Dense Master Equations ($n = 4$):** Comparison between analytical observable predictions and dense $16 \times 16$ density-matrix Lindblad simulation yielded a maximum deviation of $2.99 \times 10^{-15}$, confirming exact numerical agreement to machine precision.
2) **Forward Pauli-Frame Monte Carlo (16,384 shots):** Figure 10 displays forward Monte Carlo sampling deviations up to 32 qubits, verifying that empirical errors ($\leq 0.0043$) remain strictly inside simultaneous Hoeffding confidence bounds ($\epsilon \approx 0.033$–$0.037$ at $\alpha = 0.001$).

### H. Interpretation of the Comparative Evidence

The GP and ordinary convex formulations solve the same mathematical relaxation; their numerical agreement is a software and formulation check, not independent evidence that the physical model is accurate. The path-flow dual bracket bounds the optimum only for the declared fixed-DAG objective. The discrete-library transfer gap is sufficiently wide that it should not be read as a tight device-level performance certificate. Comparisons against the adapted slack heuristic are informative about modeled trade-offs, but unequal makespans confound unconstrained fidelity rankings. Accordingly, we treat matched-deadline results as the primary comparison and retain regressions. The 24 feasible large-circuit instances establish that the continuous solver can run at 128 qubits under attainable modeled targets, with its 24.5% makespan reduction plateau explained by boundary box saturation. They do not resolve coefficient identification, idle noise, or hardware control feasibility.

## IX. Discussion and Compiler Integration

### A. End-to-End Compiler Pass Architecture

We packaged the optimization engine as a native Qiskit compilation pass (`ConvexDurationAllocationPass`), subclassing `AnalysisPass`. The pass operates directly on the transpiler's `DAGCircuit` representation through a streamlined pipeline:

1) **DAG Extraction:** Extracts the directed acyclic dependency graph $\mathcal{D}$, identifying causal scheduling precedence between successive gates on overlapping qubit supports without changing the gate order.
2) **Analytic Infeasibility Pre-Screening:** In $O(\mathrm{nnz}(H))$ time, computes unconstrained gatewise noise minima $\tau_j^* = \mathrm{clip}(\sqrt{a_j/\gamma_j}, \ell_j, u_j)$ and evaluates $\max_k(\mu_k - \beta_k) \leq 0$ with precomputed sensitivity matrix $H$. If violated, it records an analytic infeasibility diagnostic without invoking the numerical solver.
3) **Convex Duration Allocation:** Formulates the posynomial program with safety exposure $q\gamma_j$ and invokes Clarabel/CVXPY, yielding a numerical continuous-duration solution on feasible instances.
4) **Log-Grid Quantization:** Maps continuous allocations $\tau^*$ to an abstract geometric pulse library via upward ceiling quantization with box clamping $d_j^* = \min(u_j, \ell_j q^{\lceil \log_q(\tau_j^*/\ell_j) \rceil})$, with a conditional within-model feasibility bound.
5) **Property Set Annotation:** Annotates the pass manager's `property_set` with duration vectors, modeled makespan, and any computed dual gap; the prototype does not synthesize executable pulses.

### B. Algorithmic Routine and Constrained Dual Certification

For observable-constrained duration allocation, let $w \in \mathbb{R}_+^K$ denote Lagrange multipliers for normalized observable constraints $\sum_j \frac{h_{kj}}{\beta_k}(a_j/\tau_j + q\gamma_j\tau_j) \leq 1$. Grouping gate coefficients into $\tilde{A}_j(w) = a_j \sum_k \frac{w_k h_{kj}}{\beta_k}$ and $\tilde{C}_j(w) = q\gamma_j \sum_k \frac{w_k h_{kj}}{\beta_k}$, Lagrangian relaxation with dual DAG path pressure $\rho \in \frac{1}{M}\mathrm{conv}(\mathcal{P})$ yields the constrained dual lower bound:

$$D(w,\rho) = \sum_{j=1}^{m} \min_{\ell_j \leq t \leq u_j} \left[ \frac{\tilde{A}_j(w)}{t} + (\tilde{C}_j(w) + \rho_j)t \right] - \sum_{k=1}^{K} w_k. \tag{27}$$

By weak duality, $D(w,\rho) \leq M_{\mathrm{cont}}^*/\bar{M}$ for all dual-feasible $(w,\rho)$, where $M_{\mathrm{cont}}^* = \mathrm{LongestPath}(\mathcal{D}, \tau^*)$ is the optimal makespan of the continuous relaxation. The path-flow Frank–Wolfe routine ascends on $\rho$ to compute $D(w^*,\rho^*)$. The optimality gap certifies the numerical convergence bracket of this continuous solve:

$$\epsilon_{\mathrm{gap}} = \frac{M_{\mathrm{cont}}^*/\bar{M} - D(w^*,\rho^*)}{M_{\mathrm{cont}}^*/\bar{M}}. \tag{28}$$

Subsequent discrete log-grid rounding yields discrete makespan $M_{\mathrm{disc}}^* = \mathrm{LongestPath}(\mathcal{D}, d^*)$, with guaranteed

**Algorithm 1:** Observable-Constrained Gate Duration Allocation

**Input:** Circuit DAG $\mathcal{D} = (V, E)$, duration bounds $[\ell_j, u_j]$, noise rates $a_j, \gamma_j$, sensitivity matrix $H = (h_{kj}) \in \mathbb{R}^{K\times m}$, target observable fidelities $\tau_k^{\text{target}}$, grid ratio $q > 1$, baseline makespan $\bar{M}$.
**Output:** Model-feasible discrete schedule $d^* \in \mathbb{R}^m$, discrete makespan $M^*_{\text{disc}}$, continuous relaxation makespan $M^*_{\text{cont}}$, relative optimality gap certificate $\epsilon_{\text{gap}}$, or Infeasible diagnostic.

1) Set $\beta_k \leftarrow -\frac{1}{2}\ln \tau_k^{\text{target}}$ for each target observable $k \in \{1, \dots, K\}$.
2) In $O(m)$ time, compute gatewise unconstrained minimizers $\tau_j^{\text{unc}} \leftarrow \text{clip}(\sqrt{a_j/\gamma_j}, \ell_j, u_j)$ and safety minimizers $\tau_j^{\text{safe}} \leftarrow \text{clip}(\sqrt{a_j/(q\gamma_j)}, \ell_j, u_j)$.
3) In $O(\text{nnz}(H))$ time, compute lower-bound hazards: $\mu \leftarrow H(a/\tau^{\text{unc}} + \gamma\tau^{\text{unc}})$ and $\mu^{\text{safe}} \leftarrow H(a/\tau^{\text{safe}} + q\gamma\tau^{\text{safe}})$.
4) **if** $\max_k(\mu_k - \beta_k) > 0$ **then return** Infeasible (target unachievable under baseline noise).
5) **if** $\max_k(\mu_k^{\text{safe}} - \beta_k) > 0$ **then return** Infeasible (safety margin $q$ uncertifiable).
6) Formulate convex posynomial program with safety exposure $\tilde{\gamma}_j \leftarrow q\gamma_j$ (noting that off-critical gate durations are non-unique under makespan minimization per Remark 1):

$$\begin{aligned} \min_{\tau, f, M} \quad & M \\ \text{s.t.} \quad & \sum_j h_{kj}\left(\frac{a_j}{\tau_j} + q\gamma_j\tau_j\right) \le \beta_k, \quad \forall k, \\ & f_i + \tau_j \le f_j, \ \forall (i \to j) \in \mathcal{D}, \quad \tau_i \le f_i \le M, \\ & \ell_j \le \tau_j \le u_j, \quad \forall j. \end{aligned}$$

7) Solve via Clarabel/CVXPY and check the returned primal constraint residual; reject a numerical result that exceeds the implemented feasibility tolerance. **if** solver status is infeasible **then return** Infeasible (no schedule satisfies precedence and bounds under safety margin $q$).
8) Retrieve continuous optimum $\tau^*$ and dual multipliers $w^* \ge 0$.
9) Compute continuous makespan $M^*_{\text{cont}} \leftarrow \text{LongestPath}(\mathcal{D}, \tau^*)$.
10) Compute dual lower bound $D(w^*, \rho^*)$ via (27) and evaluate continuous gap:

$$\epsilon_{\text{gap}} \leftarrow \frac{M^*_{\text{cont}}/\bar{M} - D(w^*, \rho^*)}{M^*_{\text{cont}}/\bar{M}}.$$

11) For each gate $j$, round upward with upper-bound box clamping:

$$d_j^* \leftarrow \min\left(u_j,\ \ell_j q^{\lceil \log_q(\tau_j^*/\ell_j)\rceil}\right).$$

12) Compute model-based discrete makespan $M^*_{\text{disc}} \leftarrow \text{LongestPath}(\mathcal{D}, d^*)$.
13) **return** $d^*$, $M^*_{\text{disc}}$, $M^*_{\text{cont}}$, and primal–dual gap certificate $\epsilon_{\text{gap}}$.

upper bound $M^*_{\text{disc}} \le qM^*_{\text{cont}}$. The complete routine is formalized in Algorithm 1.

### C. Scope, Missing Baselines, and Physical Validation

The geometric grid is an abstract library; it is not a measured set of alternate pulses on *ibm_torino*. A clock-aligned controller may require different rounding and an experimentally characterized noise response. The Qiskit implementation is an analysis pass that attaches duration results to a property set; it does not upload pulses or execute a modified schedule on hardware. The direct comparison uses Qiskit's published ASAP/ALAP and DD passes [17]; it holds circuit and deadline fixed, but GP has hypothetical continuous-duration actions unavailable to those passes. Its rank reversal under static detuning limits rather than establishes a superiority claim.

The largest source of uncertainty for the target device is the unmeasured split between inverse-duration and duration-proportional hazards. The snapshot determines one nominal point per gate, so multiple duration-error curves interpolate it equally well. Varying $\alpha$ probes this model uncertainty but cannot estimate $\alpha$. The external measured single-qubit sweep shows that the assumed law can miss uncertainty-scale structure even where duration is varied experimentally; its coefficients cannot be transferred to IBM CZ gates. The Heron pilot varies *idle* duration, not driven-gate duration; it cannot identify this split either. The first fixed-native-gate QPU check fails its prespecified nominal-model tolerance in every family; the second passes a six-family descriptive tolerance after a backend calibration update. Readout recalibration from a separate job did not improve the second check. Thus the fixed-gate results are calibration-sensitive and cannot establish model transfer across times, much less the duration dependence needed for GP deployment. An inverse-square candidate improves exploratory prediction error while preserving convexity, but remains inconsistent with reported measurement uncertainty and has no independent confirmation. The retrospective PCHIP decision audit improves choices in two families relative to the GP laws, but its larger-than-reported residuals and post hoc selection rule out a prospective accuracy claim. The GP proof for the evaluated circuit study still assumes isotropic post-gate Pauli noise and excludes schedule-dependent idle processes. The separate simulator probes anisotropic errors, idle $T_1/T_\phi$, and shot-static detuning during delays, but does not establish convexity for those additions. It omits detuning during driven gates, coherent crosstalk, leakage, and pulse-shape dependence. Dense Lindblad and Monte Carlo agreement only establish that separate implementations evaluate their shared assumed model consistently.

The native X experiment establishes that two supported gate durations can yield different physical returns under a matched schedule, and that a training-selected slow implementation can outperform equal-duration delayed-X and standard two-X DD controls on one held-out X-basis task. It does not show that the proposed inverse-plus-linear law chose the gate, that a GP solver improved hardware output, or that the ranking persists across bases: the Z-basis control reverses it. A gate-dependent, observable-aware model that accounts for driven evolution and coherent idle effects must be specified and fitted on one set of durations/circuits, then prospectively tested on independent choices and circuits. The existing two-point X library cannot identify a robust duration-response curve or test a continuous optimum. Wider hardware-compatible pulse libraries and matched, randomized comparisons against compiler baselines are necessary before asserting GP hardware benefit. The present contribution remains a reproducible convex timing method with conditional mathematical bounds plus a narrow physical gate-choice result.

## X. Conclusion

For a fixed circuit DAG and declared inverse-plus-linear noise model, gate-duration allocation has an exact convex representation and an inspectable path-flow lower bound. The same convex structure accommodates a declared positive inverse-power model. Under post-gate isotropic Pauli noise on Clifford circuits, stabilizer-expectation targets produce convex constraints and conditional safe rounding for the evaluated law. The model-attainable study shows feasible continuous solves through 128 qubits, while the original strict-target study mainly demonstrates fast analytic rejection. Experimental duration sweeps reveal sizeable departures from the evaluated two-parameter law; an exploratory inverse-square variant reduces prediction error but does not explain reported uncertainty. A retrospective empirical finite-library predictor improves selected measured durations in two pulse families without validating the GP error law. The direct local GP-versus-Qiskit comparison shows only small modeled Markovian gains and a substantial regression against DD under shot-static detuning. The Heron idle pilot (Appendix A) observes idle-time dependence and coherent dephasing; two fixed-native-gate checks (Appendix B) demonstrate calibration-sensitive discrepancies. A further training-selected, held-out native-X experiment demonstrates an X-basis return improvement over equal-duration delayed-X and standard two-X DD controls on one qubit, but a Z-basis reversal and the absence of a GP-selected physical schedule prevent attribution of that gain to the proposed optimizer. Broader measured duration libraries, a model that captures basis-dependent idle and driven noise, and prospective GP-versus-compiler hardware comparisons remain necessary for a GP performance claim.

## Appendix A
## Exploratory Heron Idle-Noise Pilot

We executed a small, separate timing pilot on `ibm_marrakesh` through the free Open instance (job `db4ll9o4qg6s73c2buj0`). Four one-qubit circuits on physical qubit 0 were interleaved in two shuffled blocks, with 2,048 shots per circuit. Each prepared and measured an $X$-basis state with two logical Hadamards. The three Ramsey circuits inserted a 1, 20, or 80 $\mu$s idle delay; the fourth inserted two 40 $\mu$s delays separated by $X$, followed by a second $X$ before the final Hadamard, preserving the ideal operation. Qiskit transpilation at optimization level zero retained all requested delays. The backend calibration snapshot immediately preceding the job reported $T_1 = 298.5$ $\mu$s and $T_2 = 40.4$ $\mu$s for that qubit.

Table XI gives the raw outcome fractions. Both blocks showed a higher zero frequency with the echo sequence than with the 80-$\mu$s Ramsey sequence; the blockwise differences were 0.1021 and 0.1201. This is exploratory evidence that idle dynamics and control placement matter on the target architecture. Importantly, the 80 $\mu$s Ramsey outcome $P(0) = 0.4753$ sits approximately $3\sigma$ below 0.5 (the completely mixed state), pointing to coherent phase accumulation and detuning rather than pure exponential $T_2$ dephasing. It is not a matched gate-count comparison, does not isolate a decay constant from coherent detuning and readout effects, and does not test a GP-selected gate duration. The job consumed 7 s of the free QPU allowance.

TABLE XI
Exploratory *ibm_marrakesh* idle-noise pilot on physical qubit 0: two blocks per condition, 2,048 shots per block. $P(0)$ is an observed frequency, not circuit fidelity.

| Circuit | Zero counts/shots | $P(0)$ |
|---|---|---|
| Ramsey, 1 $\mu$s | 4025/4096 | 0.9827 |
| Ramsey, 20 $\mu$s | 2655/4096 | 0.6482 |
| Ramsey, 80 $\mu$s | 1947/4096 | 0.4753 |
| Echo, 80 $\mu$s | 2402/4096 | 0.5864 |

TABLE XII
Prespecified nominal native-gate model check on *ibm_marrakesh*; each family has three blocks of 2,048 shots. Residual is observed minus pre-job predicted event probability.

| Family | Successes/6144 | Predicted | Residual | Rel. Error |
|---|---|---|---|---|
| Single return | 6119 | 0.9685 | +0.0274 | +2.8% |
| Bell parity | 6065 | 0.9619 | +0.0253 | +2.6% |
| Bell return | 6000 | 0.9512 | +0.0254 | +2.7% |
| GHZ parity | 5984 | 0.9201 | +0.0538 | +5.8% |

## Appendix B
## Prespecified Heron Native-Gate Model Checks and Relative Error Analysis

We tested whether the declared independent post-gate isotropic Pauli model, anchored to the backend's reported nominal gate errors and independent readout assignment errors, predicts outcomes for fixed native gates. Before submission, we saved a calibration snapshot and predicted distributions for four small Clifford circuit families, compiled to physical qubits 0–2 on `ibm_marrakesh`. Three shuffled blocks each contained one instance of every family, with 2,048 shots per instance (job `db4nov04qg6s73c2e8d0`). The prespecified events were return to `0`, Bell parity `00/11`, Bell return to `00`, and GHZ parity `000/111`.

All four discrepancies exceed the prespecified absolute tolerance of 0.02. Crucially, the descriptive absolute tolerance of 0.02 is loose relative to target device gate error rates (0.3%–4.8%). In relative error metrics, the model underpredicts hardware success by +2.6% to +5.8%. The pre-job snapshot's qubit-0 probability of reporting 1 after preparing 0 was 0.03027, while the return circuit produced only 25 ones in 6,144 shots. This suggests calibration drift or unmodeled state-preparation fidelity.

We then froze a separate diagnostic before collecting new data: 18 readout-calibration circuits ($|0\rangle$ and $X|0\rangle$ on each of qubits 0–2, in three randomized blocks) and 18 held-out target circuits (six families in three randomized blocks), each at 2,048 shots. The calibration job `db4o3784qg6s73c2ekqg` completed before a SHA-256 lock of both the new backend-

TABLE XIII

SECOND, HELD-OUT FIXED-NATIVE-GATE CHECK ON *IBM_MARRAKESH*: THREE 2,048-SHOT BLOCKS PER FAMILY. ENTRIES ARE TARGET-EVENT PROBABILITIES; CALIBRATED PREDICTIONS CHANGE ONLY MEASURED OPERATIONAL READOUT PARAMETERS.

| Family | Observed | Snapshot | Calibrated | Rel. Error |
|---|---|---|---|---|
| Single return | 0.9937 | 0.9948 | 0.9951 | -0.1% |
| Single-one return | 0.9919 | 0.9917 | 0.9924 | +0.0% |
| Bell parity | 0.9849 | 0.9871 | 0.9872 | -0.2% |
| Bell return | 0.9740 | 0.9825 | 0.9851 | -0.9% |
| GHZ parity | 0.9723 | 0.9728 | 0.9755 | -0.1% |
| GHZ return | 0.9587 | 0.9632 | 0.9686 | -0.5% |

snapshot and empirically readout-adjusted predictions; only then was target job `db4o3bo4qg6s73c2ekvg` submitted.

While all six readout-adjusted event residuals were below the absolute 0.02 tolerance, the model now slightly overpredicts hardware success (relative errors ranging from $-0.1\%$ to $-0.9\%$), and the raw snapshot prediction was more accurate overall than the empirical readout correction (mean absolute error 0.00286 versus 0.00476). The new backend calibration, time-stamped 02:47 local time, reported much smaller qubit-0 and qubit-2 readout errors than the earlier 02:28 snapshot (e.g., $P(1|0)$ on qubit 0 fell from 0.03027 to 0.00391). This sign reversal across consecutive jobs highlights temporal calibration drift. These fixed-gate observations neither validate the duration-error law nor demonstrate a GP schedule advantage.

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**Competing interests.** The author declares no relevant financial or non-financial competing interests.

**Data and code availability.** The source, frozen protocols, raw QPU counts and QPY circuits, analysis scripts, benchmark outputs, and manuscript source are available at https://github.com/Youssef-Ahmed38/quantum-gp-circuit-optimization and in the accompanying supplementary archive (Online Resource 1).

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